

PC-07 · ACTIVES & VITAMINS · PERSONAL CARE

HS Allantoin

Allantoin

INCI	CAS	EC	FAMILY	SUPPLIED AS
Allantoin	97-59-6	202-592-8	PC-07	Per molecule

01 POSITIONING

Each of these is a claim on the front of a pack — sourced with the purity that claim requires.

02 HOW IT WORKS

- 01 Retinol-alternative signaling

Bakuchiol mimics retinol's gene-expression profile without the irritation or photo-instability.
- 02 Sensory signaling

Menthol triggers cold receptors for an instant refreshing cue; methyl salicylate adds warming counterpoint.
- 03 Barrier soothing

Allantoin promotes keratolysis and skin-soothing, reducing the sting of active systems.

03 APPLICATIONS

- Retinol-alternative anti-aging (bakuchiol)
- Cooling after-sun and sports gels (menthol)
- Soothing and sensitive-skin lines (allantoin)
- Warming muscle and scalp products (methyl salicylate)

04 FORMULATION GUIDE

USE LEVEL	0.1–2%
PH & CONDITIONS	Molecule-dependent; bakuchiol oil-phase, allantoin water-phase ≤ 0.5% solubility
— Bakuchiol in the oil phase; allantoin dissolves slowly — respect its solubility ceiling to avoid recrystallization.	
— Menthol and methyl salicylate are volatile and potent — dose precisely and add cool.	

05 TYPICAL SPECIFICATION

Appearance	per molecule: crystals / powder / liquid
Assay	≥ 98–99%
Identity	per COA

Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.

06 PERFORMANCE – DIRECTIONAL

Irritation vs retinol (bakuchiol, class data)

Bakuchiol

Soothing effect (allantoin, class data)

With allantoin

Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.